

ECO-LOOP BUILDING AGENTS

Autonomous Closed-Loop Control for Smart Building Energy Optimization

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1. Executive Summary & Problem Background

Buildings consume approximately 40% of global primary energy and remain a primary driver of greenhouse gas emissions. Traditional Building Management Systems (BMS) operate on rigid, static schedules (e.g., fixed 23°C cooling setpoint regardless of weather, occupancy, or electricity tariff rates).

Eco-Loop Building Agents solves this operational gap by creating a live, autonomous Physical AI closed-loop control system. It pairs **EnergyPlus** (the physics simulation engine sandbox) with an **Open-Source LLM Cognitive Engine** via a standardized **Model Context Protocol (MCP)** interface. The AI agent continuously ingests real-time building telemetry, evaluates Fanger PMV thermal comfort indices, carbon intensity, and Time-of-Use electricity rates, and forward-injects dynamic thermostat setpoint overrides directly back into EnergyPlus—achieving quantifiable energy savings ($\geq 15\%$) and peak demand shaving without human intervention.

2. Tool-Calling & Model Context Protocol (MCP) Architecture

To ensure portability and avoid vendor lock-in, communication between the EnergyPlus physics engine sandbox and the LLM brain is governed by an **MCP JSON-RPC 2.0 Server** exposing standardized tools:

MCP Tool Name	Function & Description
read_building_telemetry	Streams zone temperatures, occupancy counts, HVAC power (kW), relative humidity, and PMV comfort indices.
get_energy_forecast	Fetches upcoming 4-hour forecast for ambient weather, Time-of-Use electricity tariffs (\$/kWh), and carbon intensity (gCO2/kWh).
write_thermostat_setpoints	Forward-injects dynamic heating and cooling thermostat setpoint overrides into specific zones in memory via EMS actuators.
push_ecm_action	Executes supervisory Energy Conservation Measures (PreCooling, DemandResponsePeakShaving, UnoccupiedNightSetback).
parse_idf_model	Parses EnergyPlus Input Data Files (.idf) to inspect zone geometry, envelope materials, and default schedules.
extract_simulation_logs	Queries runtime EnergyPlus simulation warning logs, severe error files, and energy summary tables.

3. Prompt Engineering Strategies & Context Latency Management

High-frequency physical simulation engines produce massive amounts of raw data. Ingesting uncompressed simulation logs directly into an LLM context window causes prompt bloat, high token latency, and reasoning degradation.

Eco-Loop implements a 3-Layer Latency Management Strategy:

- **Aggregated Telemetry Frames:** Raw sub-minute physics updates are aggregated into 15-minute supervisory telemetry frames before prompt generation.
- **Supervisory Control Horizon:** The LLM operates as a high-level supervisory manager updating hourly setpoint vectors, while EnergyPlus EMS handles rapid sub-minute setpoint holding.
- **Structured JSON Output & Guardrails:** Prompts enforce strict JSON tool-calling schemas with safety bounds (Cooling: 21.5°C–26.0°C, Heating: 18.0°C–21.5°C), preventing out-of-bounds hallucinations.

4. Technical Approach to Handling Lengthy Simulation Logs

EnergyPlus generates extensive plain-text log files (eplusout.err, eplusout.eio, eplusout.eso). Eco-Loop manages log data efficiently by:

1. Filtering runtime errors using regex pattern extractors in the extract_simulation_logs MCP tool to extract active warnings.
2. Maintaining in-memory performance accumulators for total energy (kWh), peak demand (kW), carbon emissions (kg CO2), and discomfort hours (PMV), bypassing disk read latency during runtime execution.

5. Quantitative Savings Summary & Results

Performance Metric	Baseline (Static Schedule)	Eco-Loop AI (Autonomous Agent)	Quantifiable Savings		
Total Energy Consumed	312.4 kWh	254.0 kWh	18.7% Reduction		
Peak Electrical Demand	28.5 kW	22.1 kW	22.5% Peak Shaving		
Carbon Emissions	104.2 kg CO2	81.5 kg CO2	21.8% Avoided		
Thermal Comfort (PMV)	Within [-0.5, +0.5]	Within [-0.5, +0.5]	100% Compliant	ASHRAE	55

6. Conclusion & Impact

Eco-Loop Building Agents demonstrates that combining physics-based simulation sandboxes with open-source LLMs and standardized MCP protocols successfully transforms passive buildings into active, self-correcting Physical AI structures—delivering proven energy savings while safeguarding occupant thermal comfort.